Dayeon Kang

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RESEARCH INTERESTS

My research focuses on the security and privacy of autonomous AI agents in cloud and networked systems. ***On the measurement side***, I study how AI agents behave in real-world deployments and analyze the security risks that emerge as the cloud infrastructure behind these systems scales. ***On the systems side***, I build privacy-preserving agents that leverage cloud and network infrastructure, including agent communication.

Keywords: Autonomous AI Agents, Security and Privacy on Cloud Systems, Network and Distributed Systems

EDUCATION

University of Massachusetts Amherst (UMass Amherst), MA, USA — Sep. 2023 – Present
MS/PhD Student in Computer Science

- Cumulative GPA: 3.9/4.0, as of Spring 2026
- Advisors: Prof. Amir Houmansadr, Prof. Pubali Datta

Ewha Womans University (Ewha WU), Seoul, South Korea — Mar. 2019 – Aug. 2023
B.Eng. in Computer Science and Engineering

- Cumulative GPA: 4.16/4.3, *Summa Cum Laude*

PUBLICATIONS

Dayeon Kang, J. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose Agent Are You? Multi-Layer Fingerprinting and Attribution of Autonomous Web Agents," to be submitted to ACM *The Web Conference, 2027*. [preprint]

Dayeon Kang, C. John, A. Houmansadr, E. Pauley, and P. Datta, "Characterizing Latent Configuration Security Risks in Serverless Applications," to be submitted to ACM *The Web Conference, 2027*.

Dayeon Kang, E. Yeboah-Duako, S. Thomas, and P. Datta, "When Apps Outlive Vendors: Security Implications of IoT Abandonware," ACM Conference on Computer and Communications Security *(CCS), 2026*. [paper]

Dayeon Kang, J. Sheffey, M. Wu, P. Datta, and A. Houmansadr, "CensorLess: Cost-Efficient Censorship Circumvention Through Serverless Cloud Functions," Privacy Enhancing Technologies Symposium *(PETS), 2026*. [paper]

J. Sheffey, A. Zohaib, **Dayeon Kang**, Z. Durumeric, A. Houmansadr, and Q. Wu, "I'll Shake Your Hand: What Happens After DNS Poisoning," Free and Open Communications on the Internet *(FOCI), 2025*. [paper]

Y. You, J. Yoon, **Dayeon Kang**, J. Kim, and H. Lee, "CollageMap: Tailoring Generative Fingerprint Map via Obstacle-Aware Adaptation for Site-Survey-Free Indoor Localization," IEEE Pervasive Computing and Communications *(IEEE PerCom), 2025*. [paper]

J. Yoon, Y. You, **Dayeon Kang**, J. Kim, and H. Lee, "CAMPUSRSSI: Real-World Wi-Fi RSSI Measurements for Indoor Localization," IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events *(IEEE PerCom Workshops), 2025*. [paper]

J. Yoon, Y. You, **Dayeon Kang**, J. Kim, and H. Lee, "GAN-Loc: Empowering Indoor Localization for Unknown Areas via Generative Fingerprint Map," IEEE International Conference on Sensing, Communication, and Networking *(IEEE SECON), 2024*. [paper]

Dayeon Kang, J. Kim, D. Kim, H. Kim, and K. Chae, "Grape Grade Discrimination System using Step-by-step Deep Learning Model and Drone Path Planning Algorithm," *Korea Computer Congress, 2022*.

Poster

Dayeon Kang, P. Datta, and E. Pauley, "Poster: Measuring the Risk of Exposed Configurations in Serverless Environment," IEEE Symposium on Security and Privacy *(IEEE S&P), 2026*. [poster]

RESEARCH EXPERIENCE

The Secure, Private Internet Research Group (SPIN), UMass Amherst

PhD Student (Advisor: Prof. Amir Houmansadr, Co-advisor: Prof. Pubali Datta) Sep. 2023 – Present

- Designing privacy-preserving communication for autonomous AI agents by developing a traffic-humanization framework that mitigates network-level agent fingerprinting and protects user prompt privacy across cloud and networked systems
- Fingerprinted network patterns and behavioral signals of six AI agents (AutoGen, Browser Use, Skyvern, Claude, Gemini, and ChatGPT) according to their systemic characteristics and user-generated prompts
- Measured risks of latent configurations in 16,530 serverless repositories and analyzed how configurations across different cloud providers, providing applications' structure complexity, inflate attack surfaces
- Measured and conducted meta-analysis on vulnerabilities caused by obsolete libraries, hijackable domains, and broken data flow in abandoned IoT applications
- Built serverless function-based (e.g., AWS Lambda, Microsoft Azure Functions) cost-efficient censorship circumvention proxy with Node.js; approach attracted interest from The Tor Project and is currently being applied to the communication channel of Tor's Snowflake pluggable transport

Intelligent Networked Systems Lab, Ewha WU

Undergraduate Researcher (Advisor: Prof. HyungJune Lee) Jul. 2021 – Feb. 2023

- Designed initial system to generate obstacle-aware map by reflecting degree of obstacles for each designated in-building structures
- Created Android application with Android Studio to measure Wi-Fi signal strength (e.g., RSSI)
- Implemented indoor localization purpose GAN using TensorFlow and participated in implementing federated learning model, particularly regularizing heterogeneous models (e.g., VGG, ResNet) sequentially trained on tasks

TEACHING & MENTORSHIP EXPERIENCE

Teaching Assistant (TA), UMass Amherst

- Introduction to Computer and Network Security (COMPSCI 360) Fall 2025
- System Defense and Test (COMPSCI 561) Spring 2024, 2025, 2026
- Introduction to Computer and Network Security (COMPSCI 560) Fall 2024

Peer Instructor (Tutor) in Major Courses, Ewha WU

- Computer Programming & Lab (36339-05) Spring 2022
- Computer Architecture (20493-01) Fall 2021, 2022

Mentor of Undergraduate Research Volunteers (URV), UMass Amherst Summer 2024, Winter 2025

- Instructed research projects undertaken by undergraduates and guided poster presentation

Mentor of 2022 Samsung Junior SW Creation Contest, Samsung SDS Jul. 2022 – Aug. 2022
- Served as mentor to middle school student's environment-themed software project

INVITED TALKS & SERVICES

Guest Talk | Women in Cybersecurity (WiCyS) at Umass Mar. 2025
- Journey to Security Researcher

Guest Lecture | UMass COMPSCI 560: Introduction to Computer and Network Security Dec. 2024
- Break the anonymity of Tor network: What is traffic analysis?

PETS 2027 Tadpole Reviewer

External Reviewer of USENIX Security 2026

HONORS & AWARDS

Honors

Appenzeller Memorial Scholarship, International Foundation for Ewha WU, Inc. Mar. 2026
Jung-Hun Scholarship, Jung-Hun Scholarship Foundation Mar. 2021 – Aug. 2023
Honors Scholarship (for academic excellence), Ewha WU Spring 2020, Fall 2021, Spring 2022
Alumni Scholarship, Ewha WU Spring 2020
Dean's List, seven semesters, Ewha WU 2019 – 2022

Awards

PETS 2026 Stipends, PETS 2026 May. 2026
Fourth Place in Capstone Design Project Poster Session, Ewha WU Jun. 2022
Third Place in Engineering Capstone Design Contest, Ewha WU Dec. 2021
Third Place in Smart Farm & AI Challenge, Korea Ministry of Agriculture, Food and Rural Affairs Nov. 2021
Second Place in Software Entrepreneur Contest, Ewha WU Nov. 2021

TECHNICAL SKILLS

Proficient	C, Python (with PyTorch, TensorFlow), JavaScript (JS), Docker, MATLAB, LaTeX
Intermediate	MySQL, JAVA, C#
Basic	Assembly Language (MIPS), Android Studio